Fragment-Mixed Continuation for Cross-Corpus Visual Speech Recognition

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Abstract

Improving a visual speech recognizer on its training corpus need not improve transcription of an unfamiliar video collection. We study temporal supervision granularity during source-only continuation, preserving the architecture and whole-utterance inference. Fragment-mixed continuation alternates complete utterances with short, transcript-labelled views of the same videos. Comparisons match initialization, source-clip order and 6,000 optimizer updates. On 1,167 ViCocktail clips after VLR continuation, fixed-time mixing reduces common-decoder word error rate (WER) by 3.41 and 3.17 percentage points across two seeds. A separate matched comparison obtains 67.37% WER with full-only continuation, 63.85% with fixed-time mixing and 63.39% with whole-word mixing. Replacing the scheduled full views with additional whole-word fragments raises destination WER to 70.72% and source WER to 44.84%, compared with 38.61% source WER for whole-word mixing. Increased deletions outweigh fewer substitutions and insertions. Retaining full-utterance presentations therefore benefits both populations in this fixed-budget comparison. Relative to full-only continuation, the two mixed states still incur source costs of 0.55 and 0.58 WER points. The fixed-time transfer gain persists at 3.46 points after component-wise token normalization. In LRS2-to-WildVSR transfer, mixing gives up 1.46 source WER points without a clear destination gain. These results support mixed temporal supervision in the studied Vietnamese continuation setting, while showing that its value depends on the training regime and must be assessed jointly on source and destination.

Keywords: Visual speech recognition, Lip reading, Cross-corpus transfer, Fragment-mixed continuation, Temporal supervision

1. Introduction

Visual speech recognition (VSR) recovers linguistic content from visible mouth movements. Large-scale supervised learning has substantially improved benchmark accuracy [1, 2], yet strong performance on a training corpus does not ensure reliable transcription of a different video collection [3]. For an existing recognizer, a practical question is how to reuse labelled source videos before deployment elsewhere. We ask whether changing the temporal scope of supervision during source-only continuation improves complete destination transcripts, and what it gives up on the source corpus.

Complete utterances supply extensive visual and linguistic context. Short views contain fewer neighboring frames and shorter target sequences, changing both the available evidence and the optimization problem. A temporal cut can also remove visual evidence for a boundary word retained in the target. These changes make short-view supervision a testable intervention, rather than an assumed improvement over complete clips.

Our initial formulation, fragment-mixed continuation (FM), alternates complete source utterances with fixed-duration, transcript-labelled fragments of the same videos. Fragment targets contain the transcript units overlapping each interval. Only the training presentations change: the recognizer keeps its architecture and ordinary whole-utterance decoding. No fragment selection, timestamp estimation or prediction merging is required at inference, and no additional source videos are introduced.

Full-clip continuation (FC) provides the comparison under the same initialization, source-clip order and 6,000 optimizer updates. A common decoder compares fitted states; development-selected decoders additionally describe calibrated systems. Equal numbers of updates and presentations do not imply equal frame or token exposure. We measure these exposures and retain the native objective, treating the recipe as a complete intervention under an update budget rather than attributing its effect to one regularization mechanism.

Two corpus pairs expose different outcomes. In VLR-to-ViCocktail transfer, FM reduces common-decoder WER by 3.41 and 3.17 percentage points across two seeds. In LRS2-to-WildVSR transfer, it gives no clear aggregate gain. The seed-17 source assessment shows that FM improves over each initializer but retains less of FC's source improvement in

both settings. The Vietnamese transfer gain thus accompanies a source cost. Differences in initialization, training coverage, repetition and calibration prevent attributing the contrast between corpus pairs to language alone.

Temporal cropping and mixed-length training are established in lip reading [4, 5]. A separate matched comparison therefore includes conventional whole-word mixing, which improves transfer and outperforms fixed-time mixing by 0.459 WER points. A paired token-normalized loss control retains the fixed-time gain. The whole-word short-only control further tests the contribution of scheduled full-clip presentations, using the same source order and unchanged whole-utterance assessment. Together, these controls test how temporal schedule, fragment labelling and loss reduction shape source retention and destination recognition. The short-only state is worse than whole-word mixing by 7.328 destination WER points and 6.238 source points, with substantially more deletions. Retaining scheduled full views is therefore beneficial in this comparison; increasing the proportion of short supervision does not monotonically improve transfer.

Two same-target controls restore neighboring pixels or supply predictions informed by the complete video. Neither improves on FM in the Vietnamese comparison. Their edit-error profiles show that improved visual coverage or fewer substitutions can be offset by additional deletions, motivating assessment of the complete output sequence.

The paper contributes three results:

- A controlled study of source-only short-view continuation under matched update budgets, with unchanged whole-utterance recognition at deployment.
- A decoder-controlled comparison across two corpus pairs, including a two-seed Vietnamese transfer gain and joint source/destination assessment that identifies where the intervention does not improve recognition.
- Comparisons of temporal schedule, crop labelling, loss reduction and same-target context, supported by edit-error and exposure measurements.

Together, these results characterize when mixed temporal supervision improves transfer, how retaining full utterances contributes, and the source-retention cost of the resulting continuation strategy.

2. Related work

2.1. Learning visual speech representations

Continuous lip reading combines spatial analysis of the mouth with temporal and linguistic modeling. Sentence-level systems established the importance of learning over variable-length visual sequences [6, 7]. Large-scale supervised recognition subsequently expanded the available training resources [1]. Auto-AVSR uses automatically generated labels to exploit additional audiovisual collections [2], while AV-HuBERT learns audiovisual representations through masked cluster prediction [8]. These approaches address representation quality and scale. Our experiments instead start from existing language-specific recognizers and study how the presentation of supervision affects transfer under a fixed update budget, using the same visual backbone and source-video collection.

The recognizers considered here use a hybrid CTC/attention formulation [9, 10]. The two output components provide useful controls because changing their relative decoding weight can change substitutions, deletions and insertions without changing the trained model. Accordingly, we report both a decoder selected for each checkpoint on development data and a common decoder. Together, these comparisons distinguish the performance of the tuned system from the effect observed under a shared decoding rule.

2.2. Temporal crops and short-sequence supervision

Training on shorter sequences is not new to lip reading. Early sentence-level work used a progression in sequence difficulty [6]. Martinez et al. vary temporal context around an intact target word, improving robustness to frame removal in isolated-word recognition [4]. Their BGRU comparison also illustrates that robustness and unperturbed accuracy can change in different directions. Prajwal et al. train with consecutive-word subsequences and compare increasing subsequence length with using different lengths together [5]. Their study is particularly relevant because it shows that mixed temporal scales provide an

established augmentation baseline. We build on this setting to examine transfer under matched continuation budgets and whole-utterance evaluation on a different corpus.

Recent word-level work also examines temporal information and augmentation. InfoSyncNet combines temporal modeling with time masking, mixup and word-boundary indicators, reporting different time-masking effects on LRW and LRW1000 [11]. These experiments retain an isolated-word recognition task. FM instead changes both the visual interval and its transcript during continuation, then evaluates complete utterances on a destination corpus.

The specific setting examined here differs in two respects. The fragment is defined by a fixed time interval rather than a fixed number of complete words. Its label contains any transcript unit that overlaps the interval, so boundary words can remain in the target despite incomplete visual coverage. Moreover, evaluation uses complete utterances from a different corpus, rather than the same fragment task used during continuation. These choices create a testable question about the interaction between context availability and transfer. We examine this fixed-time formulation through paired context-restoration controls and a direct comparison with conventional word-aligned fragments. The latter tests the crop choice within our continuation protocol; it is not a reimplementation of Prajwal et al.'s architecture or complete training pipeline. The distinction from prior work lies in the joint assessment of whole-utterance transfer, source performance and decoder dependence after continuation, rather than in temporal cropping alone.

2.3. Contextual targets and consistency learning

Self-distillation provides a way to use information that is available during training but absent from a particular input view. For example, data2vec predicts contextual representations of the full input from a masked view [12]. Closer to the recognition objective, CR-CTC encourages agreement between framewise CTC distributions from two augmented speech views [13]. Its branches share transcript supervision, with bidirectional consistency and stop-gradient targets at corresponding time positions.

Our full-view control examines broader temporal context during fragment learning. The current model supplies a stop-gradient CTC target from the complete video, restricted

to the fragment's time positions, while the student retains its fragment transcript. It therefore tests contextual guidance alongside the alternative of exposing additional context directly to the encoder. Both controls retain ordinary whole-utterance inference. This framing connects the experiments to established consistency learning while focusing the comparison on the context available when learning from partial speech.

2.4. Cross-corpus evaluation and multilingual evidence

Speaker adaptation addresses distribution shift using examples from the intended users. Kim et al. learn prompts from target-speaker adaptation data while preserving the pretrained recognizer parameters [14]. Here, recognizer gradient updates use the continuation source; destination development data are used only for decoder calibration where the protocol permits it. This distinction separates source continuation from target-speaker fitting.

WildVSR was introduced to examine whether strong VSR models generalize beyond LRS3 [3]. Its results motivate retaining an external-corpus assessment even when source-domain validation appears satisfactory. We use WildVSR as the English destination corpus and ViCocktail as the Vietnamese destination [15]. The Vietnamese source is the VLR dataset [16]; English continuation uses prepared LRS2 training clips. The two comparisons share the intervention and whole-utterance objective, but they are distinct transfer settings with different initializations, tokenizers, training coverage and development-data availability.

We analyze each corpus pair separately, examining training coverage, edit types and sequence length as possible explanations for their contrasting outcomes.

3. Fragment-mixed continuation

3.1. Task and notation

Let $X = (x_1, \ldots, x_T)$ be the prepared mouth-image sequence of a source clip, with transcript y. A visual encoder f_θ produces a sequence $H = f_\theta(X)$, and CTC and attention heads define the recognition objectives. The models used here preserve the temporal frame

count through the encoder, allowing a frame interval to index the corresponding encoded sequence. At assessment, the desired output is the complete transcript

$$\hat{y} = D_\alpha\Big(f_\theta(X)\Big), \tag{1}$$

where D_α is the selected decoder. No target interval, reference transcript, word timestamp or crop decision is supplied during this inference.

For training only, let $I = [a, b)$ denote a fragment of $L = 50$ frames, or two seconds at 25 frames per second. The available word timings define its label as

$$y_I = \text{concat}\{w_k : e_k > a/25 \text{ and } s_k < b/25\}, \tag{2}$$

for positive-duration words with start and end times (s_k, e_k). Zero-duration entries are included only when their timestamp lies in the half-open interval. The rule is based on overlap, not on whether the word is fully visible. For example, a word ending just inside the fragment can be included even if most of its articulation lies before a. Crop locations are drawn from precomputed eligible starts, using the clip identifier, presentation index and random seed. For a clip of T frames, starts are enumerated at one-frame spacing from 16 through $T - 66$ inclusive, retaining nonempty labels with a feasible CTC path. A deterministic hash of the seed, clip identifier and presentation index selects one retained start. This supplies 16 frames on either side for the paired context control, even when FM itself uses only the core.

3.2. Full-clip and fragment-mixed supervision

Full-clip continuation (FC; experiment T0) minimizes the recognizer's existing hybrid loss on each complete source clip:

$$\mathcal{L}_{\text{ASR}}(H, y) = 0.1\,\mathcal{L}_{\text{CTC}}(H, y) + 0.9\,\mathcal{L}_{\text{att}}(H, y). \tag{3}$$

Fragment-mixed continuation (FM; T1) alternates full and fragment presentations:

$$\mathcal{L}_{\mathrm{FM}}^{(j)} = \begin{cases} \mathcal{L}_{\mathrm{ASR}}(f_\theta(X), y), & j \text{ even}, \\ \mathcal{L}_{\mathrm{ASR}}(f_\theta(X_I), y_I), & j \text{ odd}. \end{cases} \tag{4}$$

Here j indexes individual training presentations before gradient accumulation. The same source clips occur in the same order for FC and FM within a seed. Consequently, FM does not add previously unseen videos: it changes which part of an already selected video is presented. At 6,000 optimizer updates with eight presentations per update, each run contains 48,000 presentations. FM divides these equally between complete clips and fragments.

The architecture is unchanged. The visual encoder and both recognition heads are updated, while batch-normalization running statistics remain fixed. The native loss reductions are retained: CTC is summed over the example and the attention criterion aggregates valid token positions, with batch-size reduction rather than explicit token-length normalization. Thus, matching updates does not match frames, tokens or gradient magnitudes. We measure the first two directly in Section 5.7; no proportional relationship between token count and the final optimizer update is assumed.

3.3. Conventional whole-word mixing

To compare fragment labelling rules, we also construct short views containing consecutive complete word annotations. For each scheduled fixed-time interval, we enumerate annotation-bounded spans overlapping that interval, round their start down and end up to frame indices, and retain lengths of 50–82 frames (2.00–3.28 seconds). Among spans with nonempty, CTC-feasible labels, we choose the one closest to the original interval's two boundaries. Ties are resolved by proximity to 50 frames, midpoint distance and temporal order. The label contains the consecutive words defining that span. Completeness refers to the available annotations, not independently verified articulation boundaries.

This conventional alternative retains the alternating full/short schedule and unchanged whole-utterance inference. When no admissible word span exists, both the whole-word

and fixed-time arms use the complete clip at that presentation. Sharing these fallback positions prevents a difference in example coverage from being mistaken for a crop effect. The comparison changes crop extent and target content together; it does not hold input frames or target tokens exactly equal.

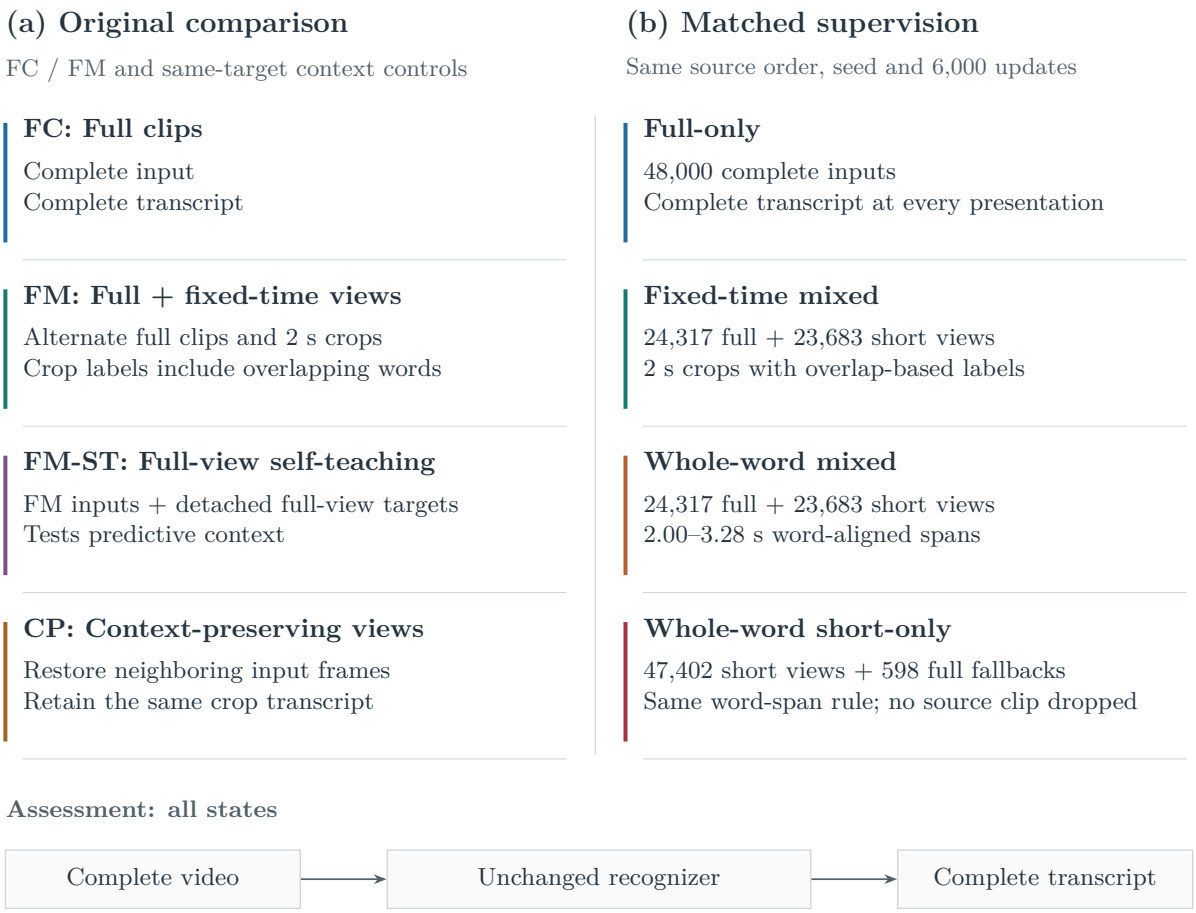


Figure 1: Training interventions with unchanged whole-utterance inference. (a) The original comparison tests full-clip continuation (FC), fragment mixing (FM), full-view self-teaching (FM-ST/T2), and restored visual context (CP/T4). (b) The matched comparison includes full-only, fixed-time mixed, whole-word mixed and whole-word short-only supervision. The mixed states share fallback positions; the short-only state also replaces the scheduled full views when an admissible word span exists. The two panels represent distinct fitted-state comparisons; all variants recognize the complete video at assessment. Schematic only.

3.4. Whole-word short-only supervision

To test whether scheduled full-utterance supervision is needed alongside short views, the short-only control preserves the whole-word mixed arm's source order and all of its odd-indexed short views. At each formerly full, even-indexed presentation, it applies the same deterministic fixed-time anchor and the whole-word span rule in Section 3.3. The

target contains only the words defining the selected span. When no admissible span exists, the complete clip and transcript are retained; no source clip is removed or replaced.

This produces 47,402 short views and 598 full-clip fallbacks in 48,000 presentations. Of the fallbacks, 317 are the positions already shared by the two mixed states and 281 occur at formerly full positions. Thus, "short-only" names the intended supervision schedule, not the absence of every full-clip example. The actual exposure is 2,719,321 input frames and 422,153 target tokens, with 470,153 valid attention positions including end-of-sequence tokens. The native recognition objective, initializer, source order, optimizer schedule and final 6,000-update endpoint remain the same as in the matched mixed comparison. This control changes the balance of full and short supervision; it does not equalize frame exposure, target-token exposure or gradient magnitudes.

3.5. Controlling loss reduction

Short views change both the visual evidence and the number of supervised tokens. We therefore compare native reduction with a paired, per-example normalized objective. For a target of N native tokens, the control minimizes

$$\mathcal{L}_{\text{norm}}(H, y) = 0.1\frac{\mathcal{L}_{\text{CTC}}^{\text{sum}}(H, y)}{N} + 0.9\frac{\mathcal{L}_{\text{att}}^{\text{sum}}(H, y)}{N + 1}. \tag{5}$$

The attention term retains the recognizer's native criterion; its denominator counts valid target positions including the end-of-sequence token. The same reduction applies to full and fragment presentations. This changes the weighting of examples of different lengths and the relative contributions of the two loss components. It does not match gradient norms or optimizer trajectories. The comparison asks whether the full-versus-fragment transfer contrast persists under this explicit reduction change.

3.6. Controls for the role of context

Two variants examine whether the information removed by cropping changes the effect. Both retain the same full/fragment schedule, fragment labels, source clips and final update count. They modify only the fragment presentations described below. Their computational costs need not equal that of FM.

Full-view self-teaching (FM-ST; T2). The student processes X_I as in FM. A second forward pass processes the complete clip with the current model in evaluation mode and without gradients. Its encoded features are restricted to I before forming the teacher's CTC distribution. Let z_t^I be the student's CTC logits and z_t^F the corresponding full-view logits. With temperature $\tau = 2$, the fragment objective is

$$p_t^F = \mathrm{sg}\Big[\mathrm{softmax}(z_t^F/\tau)\Big]\,, \tag{6}$$

$$\mathcal{L}_{\text{FM-ST}} = \mathcal{L}_{\text{ASR}}(f_\theta(X_I), y_I) + \lambda \frac{\tau^2}{L} \sum_{t=1}^{L} \mathrm{KL}\Big(p_t^F \parallel \mathrm{softmax}(z_t^I/\tau)\Big)\,, \tag{7}$$

where $\lambda = 0.15$. The teacher shares the current parameters with the student; it is not an independently trained or exponential-moving-average model. Evaluation mode removes training-mode stochasticity from the teacher pass. This control asks whether a distribution informed by the complete video helps resolve the fragment's reduced evidence. At test time the auxiliary pass and loss are absent.

Context-preserving continuation (CP; T4). The input to the encoder is expanded to $J = [a - 16, b + 16)$, retaining 0.64 seconds on each side of the two-second fragment. Only the encoded positions corresponding to I are passed to the supervised recognition heads:

$$\mathcal{L}_{\text{CP}} = \mathcal{L}_{\text{ASR}}([f_\theta(X_J)]_{16:16+L}, y_I)\,. \tag{8}$$

The target label and core pixels are identical to the corresponding FM presentation. Neighboring pixels can affect the core representation through the encoder, but neighboring words are not added to the supervised target merely because context is visible. This separates retaining visual context from restoring the complete transcript. It also differs from inference-time temporal projection: no predicted word is selected by a timestamp at assessment.

3.7. Comparison structure

FC versus FM measures the net effect of replacing half the full presentations with fragments. Whole-word mixing tests a conventional alternative to the fixed-time label

rule. Whole-word mixed versus short-only supervision tests the contribution of scheduled full-utterance examples under the same short-view rule. CP and FM-ST restore context through pixels and predictive distributions, respectively, while keeping fragment targets fixed. Because an overlap-selected label can extend beyond the crop, we report annotation coverage alongside training exposure and complete-transcript recognition error.

4. Experimental design

4.1. Corpora and recognition systems

We evaluate the same continuation intervention in two corpus-transfer settings: VLR to ViCocktail and LRS2 to WildVSR. Each setting has matched within-setting FC/FM comparisons and two same-target context controls. Differences between the corpus pairs include language, initialization, training coverage and development-data availability; their comparison assesses portability rather than isolating a single factor.

The Vietnamese setting continues a VLR recognizer on VLR training clips and evaluates whole-utterance transfer to ViCocktail. VLR contains Vietnamese mouth-video segments and associated text [16]; ViCocktail is a separately constructed Vietnamese audiovisual corpus [15]. The assessment covers 1,167 official ViCocktail test clips, containing 30,091 whitespace-delimited reference units and 20 original-video groups. These groups are recording-source identifiers, not verified speaker identities.

The English setting starts from an LRS3-initialized model, continues on prepared LRS2 training clips, and evaluates all 2,854 released WildVSR clips, with 48,037 reference words [7, 17, 3]. The source and destination terminology refers to continuation and assessment corpora; the initializer can have its own upstream training history. We retain a larger-data Auto-AVSR model as a practical English reference. It was trained with LRS2, LRS3, VoxCeleb2 and AVSpeech, and is not an equal-resource continuation arm.

Both continuation settings use the Auto-AVSR graph [2]: a 3D convolution followed by a framewise ResNet-18, a 12-block Conformer encoder with 768-dimensional representations, 12 attention heads and 3,072-dimensional feed-forward layers, and a six-block Transformer decoder of the same width. The Conformer convolution kernel has length 31.

Table 1: Data and exposure in the two continuation settings. Pool sizes refer to the prepared inputs for this study, not the total size of the original datasets.

Quantity	Vietnamese	English
Prepared training clips	92,632	45,838
Fragment-eligible training clips	90,519	7,291
Distinct clips seen per matched run	48,000	7,291
Training presentations per run	48,000	48,000
Development clips for decoder selection	768	1,082
Assessment clips	1,167	2,854
Assessment reference units	30,091	48,037
Seeds in completed FC/FM comparison	17, 29	17

Vietnamese decoder selection uses ViCocktail development data; English selection uses LRS2 source-development data. Vietnamese whitespace units and English words are evaluated separately, without pooling their denominators.

The Vietnamese initializer is the VLR checkpoint; the English initializer is the released LRS3-trained `vsr_trlrs3_base` checkpoint. Model identities and tokenizer mappings are provided with the numerical supplement.

VLR supplies word timings in its annotation files. LRS2 main transcripts are preserved, with missing word boundaries derived by audio CTC forced alignment. Timings select training labels only; neither audio nor timings are supplied to the visual recognizer at inference. Both sets of training timings are automatic.

Both settings use video alone at inference. The prepared grayscale mouth regions have size 96×96; training center-crops them to 88×88 and normalizes intensities with mean 0.421 and standard deviation 0.165 after scaling to $[0, 1]$. The continuation arms within a language share preprocessing, architecture and tokenizer. The Vietnamese and English output inventories have 2,002 and 5,049 classes, respectively, including the required special symbols. Figure 2 illustrates the prepared mouth-region input in each corpus, with source and destination roles defined by the transfer comparison.

A source-performance follow-up evaluates the initializer, FC and FM at seed 17 on all 11,307 prepared VLR test clips (225,960 reference units and 61 archived source groups) and all 1,243 LRS2 test clips (6,660 words and 354 source groups). Each corpus uses the same fitted states, preprocessing and frozen readouts as its destination comparison. Source and destination results are analyzed separately; the source follow-up does not add a second training seed.

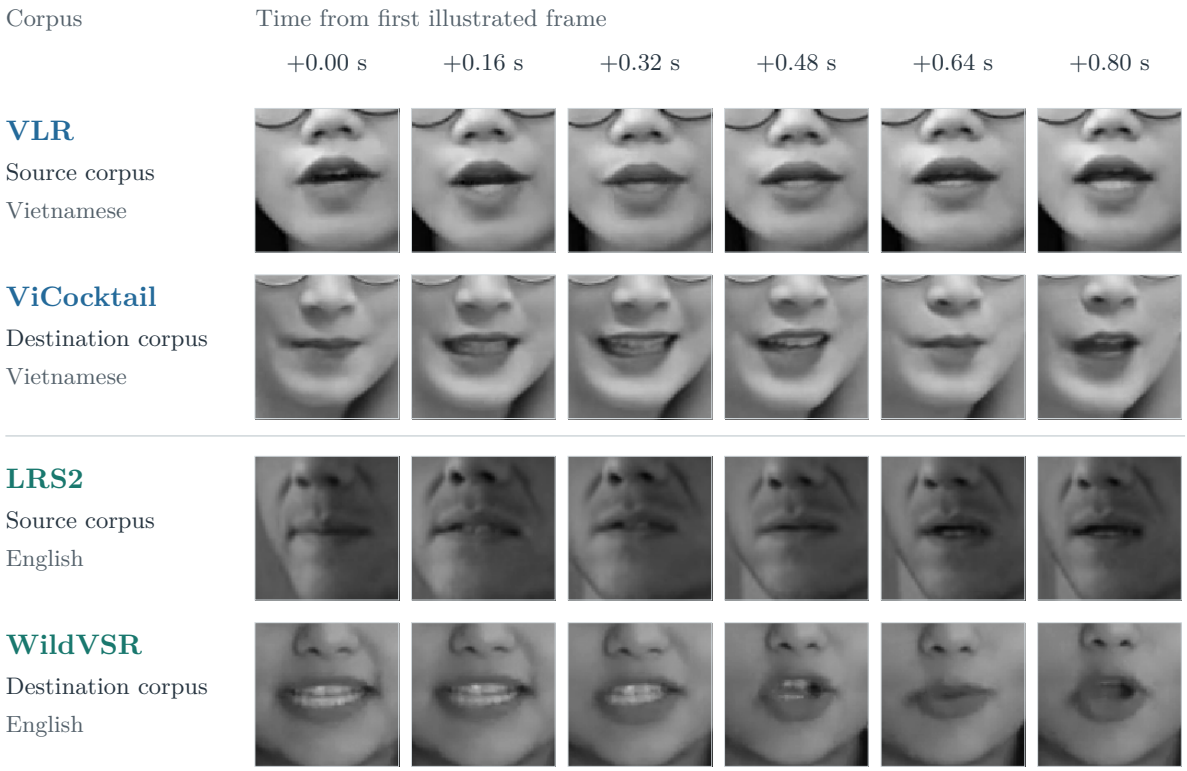


Figure 2: Prepared mouth-region sequences from VLR [16], ViCocktail [15], LRS2 [7] and WildVSR [3]. Each row shows six real grayscale 88×88 center crops at 0.16-second intervals from one clip; displayed times are relative to the first illustrated frame. These examples show the visual input, while recognition is evaluated on complete clips.

4.2. Continuation budget and controls

All matched FC/FM comparisons use the final checkpoint after 6,000 updates, rather than independently selected intermediate steps. The FM-ST and CP controls also use step 6,000 and are evaluated at seed 17 in both corpus pairs. Training uses AdamW with peak learning rate 2×10^{-6}, weight decay 0.01, 500 warm-up steps and cosine decay. Eight individually processed examples are accumulated per optimizer step; the gradient norm is clipped at 5. Training uses bfloat16 where supported, while the assessment decoder uses float32. The same seed determines the source-clip order and fragment choices across paired methods.

The fragment-eligibility rule requires 82–400 prepared frames, usable timing and feasible target sequences. This has different consequences across corpora. The Vietnamese matched runs draw 48,000 distinct clips from a pool of 90,519. The English runs draw from 7,291 eligible clips, each revisited six or seven times. Thus, both English FC and FM share the long-clip subset; FC does not receive the short clips excluded from that matched

comparison. An additional English full-pool FC run (F0) uses the complete eligible full-clip pool of 45,838 clips at the same 6,000-update budget. It tests a practical alternative training population, rather than isolating the crop transformation on identical examples.

We report the uncontinued initializer in each setting. We also report a native ViCocktail model and the larger-data English Auto-AVSR model to indicate the remaining performance gap. These practical references have different training resources; the native ViCocktail system also has its own preprocessing. They are not used to estimate the effect of FM relative to FC.

4.3. Matched comparison of fragment labelling rules

A follow-up VLR-to-ViCocktail experiment trains full-only, fixed-time and whole-word continuations together in one training runtime at seed 17. All three start from the same initializer and use the same 48,000 source-clip presentations, 6,000 updates, optimizer schedule and native hybrid objective. The two mixed arms share 317 full-clip fallback positions, yielding 24,317 full and 23,683 short presentations per arm. They are newly fitted states, distinct from the original FC/FM checkpoints. Their own source assessment covers all 11,307 VLR test clips under the same common decoder, with paired intervals resampling the 61 source-video groups. Each new state is therefore assessed on both source and destination without changing its weights or readout.

The crop rules in Section 3.3 and the final-update endpoint are fixed before this follow-up's assessment. Its main comparison uses the common decoder on the same 1,167 ViCocktail clips, without choosing a separate decoder for each new model. This is a focused augmentation control on a previously used assessment set, not a new independent corpus confirmation. Its conditional intervals use the same 20 source groups and 10,000 paired resamples. We report this comparison separately so that the shared fallbacks and fitted states are not conflated with the original two-seed experiment.

A further pre-specified short-only state at seed 17 uses the schedule in Section 3.4; the three completed native states remain its fixed comparators. Its assessment protocol covers the same 11,307 VLR source clips and 1,167 ViCocktail destination clips, using the common decoder without a new decoder search. The primary contrast is whole-word mixed

minus short-only WER; the reference contrast is full-only minus short-only WER. Positive differences favor short-only supervision. Fixed-time mixing is a supporting comparison, rather than an additional primary contrast. The two population-specific analyses use 10,000 paired resamples of the 61 VLR or 20 ViCocktail archived source groups. The plan and contrast directions were fixed before short-only training. Intervals are conditional on these fitted states, pointwise and unadjusted for multiplicity. No equivalence margin was specified, so an interval spanning zero does not establish that mixing and short-only supervision are equivalent. This follow-up also reuses previously observed assessment corpora and adds no new training seed or independent test collection.

The loss-reduction control fits a further full-only/fixed-time pair using Equation 5, with the same initializer, source schedules, shared fallbacks, seed, optimizer and final-update endpoint as the matched crop comparison. All four native/normalized states use the common decoder. We compute the difference between their full-minus-fixed-time gains using paired resamples of the same recording groups across all four states. No normalized whole-word arm is included in this comparison.

4.4. Decoder selection and metrics

For the original FC/FM comparisons and same-target context controls, each checkpoint's primary decoder is selected by aggregate error count on development data from a fixed set of 11 candidates. The matched crop, short-only and loss-reduction follow-ups use the common decoder without a new search. The candidates comprise hybrid beam decoding with CTC weights $\alpha \in \{0.1, 0.5, 0.9, 1.0\}$ and greedy CTC with blank-logit biases $b \in \{0, -0.5, 0.5, -1, 1, -2, 2\}$. Ties use a fixed candidate order. Beam decoding uses width 40, zero length penalty and no external language model. The search horizon covers the full encoded frame sequence. The common decoder is the same beam decoder with $\alpha = 0.1$ for all checkpoints.

Vietnamese primary selection uses 768 ViCocktail development clips, whereas English selection uses 1,082 LRS2 source-development clips. Vietnamese therefore allows target-domain decoder calibration without target-domain gradient updates; English uses source-only decoder calibration. We do not treat these as identical zero-shot protocols. The

common-decoder comparison removes model-specific decoder selection, although it cannot remove differences in the corpora or initializers.

Text is normalized with Unicode NFC, lowercasing and whitespace normalization. We compute corpus-level WER as $100(S+D+I)/N$, where N is the total reference unit count. Vietnamese whitespace-delimited units need not coincide with lexical words; consequently, its WER is not pooled with English WER. We report absolute percentage-point differences and the underlying substitution, deletion and insertion counts. Positive FC–FM differences favor FM.

Paired bootstrap intervals use 10,000 resamples. Vietnamese resampling preserves the 20 original-video groups; WildVSR resampling uses released clips because a validated recording-group reconstruction is unavailable. The WildVSR intervals are therefore conditional clip-resampling summaries and do not establish uncertainty over independent recording sources. The source follow-up resamples 61 VLR or 354 LRS2 archived source groups, which are not asserted to be verified speaker identities. These resampling units differ across corpora. All intervals condition on the fitted seeds and chosen decoders. For the mechanism comparison, FM–FM-ST and FM–CP are the primary contrasts; comparisons with FC describe how much of the original transfer gain remains. Intervals are pointwise, without adjustment for multiple comparisons. The assessment sets have appeared in earlier experiments; the results should therefore be interpreted as retrospective evidence rather than confirmation on an untouched test set. The source-performance follow-up also uses previously observed test data. Upstream training overlap is not independently established.

5. Results

5.1. Transfer under a common decoding rule

The two corpus pairs produce different outcomes under the same within-setting comparison. FM improves VLR-to-ViCocktail transfer by 3.410 and 3.170 WER points at seeds 17 and 29. It produces no clear LRS2-to-WildVSR improvement: the common-decoder FC–FM difference is -0.067 points, with an interval that includes zero. Figure 3 and

Table 2: Whole-utterance transfer from VLR to ViCocktail. WER (%) is lower when better.

Model	Seed	Primary	Common	Primary decoder
Initial	–	70.057	70.034	Greedy $b = -1$
FC	17	67.445	67.326	Greedy $b = -1$
FM	17	61.882	63.916	$\alpha = 0.5$
FM-ST	17	62.002	63.919	$\alpha = 0.9$
CP	17	62.444	64.793	$\alpha = 0.5$
FC	29	66.983	67.233	Greedy $b = -1$
FM	29	63.448	64.062	Greedy $b = -1$
Native ViCocktail	–	38.503	38.503	$\alpha = 0.1$

Primary decoders use 768 ViCocktail development clips. The common decoder has CTC weight 0.1 and beam size 40. FM-ST and CP have seed 17 only. The native reference uses different training data and preprocessing.

Table 3 summarize these contrasts. We first examine destination transfer and the corresponding source assessment, then analyze error changes, training exposure and contextual information.

5.1.1. VLR to ViCocktail

Table 2 presents the VLR-to-ViCocktail comparison. With the common beam decoder, replacing half the full-clip presentations with fragments reduces WER from 67.326 to 63.916 at seed 17 and from 67.233 to 64.062 at seed 29. The gains of 3.410 and 3.170 percentage points therefore persist with the same recognizer architecture, source-clip schedule, update count and decoding rule. They measure the benefit of the training intervention under this continuation budget, rather than a benefit from model-specific decoder selection.

Development-selected decoding strengthens the seed-17 result: FC/FM obtain 67.445/61.882 WER, a 5.563-point gain. At seed 29, the corresponding values are 66.983/63.448, a 3.536-point gain. The interaction with decoding affects the magnitude, but not the direction, of the Vietnamese result. FM also improves on the uncontinued initializer, which obtains 70.057 primary and 70.034 common-decoder WER. The native ViCocktail reference reaches 38.503 WER with different training resources and preprocessing. FM thus improves cross-corpus continuation while leaving a substantial gap to a system developed for the destination corpus.

Both seeds improve in all 20 original-video groups under either decoding regime. The common-decoder source-bootstrap 95% intervals are [2.892, 4.119] and [2.683, 3.885] points

Table 3: Paired FC–FM improvements and 95% bootstrap intervals (percentage points).

Transfer	Seed	Regime	Gain	95% interval
VI	17	Primary	5.563	[4.940, 6.351]
VI	29	Primary	3.536	[2.360, 4.768]
VI	17	Common	3.410	[2.892, 4.119]
VI	29	Common	3.170	[2.683, 3.885]
EN	17	Primary	0.025	[-0.371, 0.428]
EN	17	Common	-0.067	[-0.338, 0.210]

Positive values favor FM. Vietnamese intervals resample original-video groups; English intervals resample released clips. Intervals condition on the fitted models and decoders.

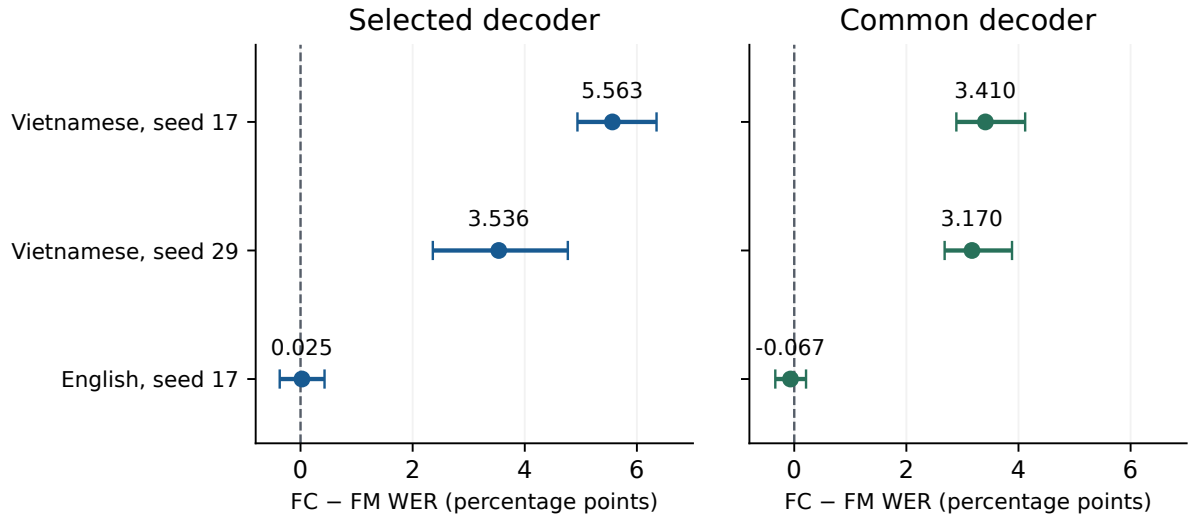


Figure 3: Paired FC–FM gains in whole-utterance WER. Positive values favor FM. Error bars are 95% bootstrap intervals. Vietnamese intervals resample original videos; English intervals resample released clips. The English result provides a cross-setting check, without pooling the two languages.

(Table 3; Figure 3). This consistency supports a gain within the studied corpus pair. The intervals summarize variation across observed recording groups conditional on the fitted models; they do not estimate variability across independently trained seeds.

5.1.2. LRS2 to WildVSR

The English experiment checks the same recipe in a different continuation setting. It does not reproduce the Vietnamese advantage (Table 4). Primary FC/FM WER is 66.817/66.792: a 0.025-point gain with a conditional clip-bootstrap interval of [−0.371, 0.428]. Under the common decoder, FM is slightly worse, 66.884 versus 66.817, with an FC–FM interval of [−0.338, 0.210]. The initializer obtains 66.164 primary and 67.092 common WER. FM is therefore descriptively worse than the initializer under selected

Table 4: Whole-utterance transfer to WildVSR. WER (%) on the same 2,854 assessment clips.

Model	Seed	Primary	Common	Primary decoder
Initial	–	66.164	67.092	$\alpha = 0.5$
FC	17	66.817	66.817	$\alpha = 0.1$
FM	17	66.792	66.884	$\alpha = 0.5$
FM-ST	17	66.771	66.894	$\alpha = 0.5$
CP	17	66.749	66.794	$\alpha = 0.5$
Full-pool FC	17	67.438	67.438	$\alpha = 0.1$
Large-data Auto-AVSR	–	36.010	36.010	$\alpha = 0.1$

Primary decoders use 1,082 LRS2 source-development clips. The common decoder has CTC weight 0.1 and beam size 40. All continuation variants use seed 17. Full-pool FC changes the training population; the large-data reference uses different pretraining resources.

decoding, but slightly better under common decoding; continuation is not uniformly helpful.

The training regimes differ materially. English fragment eligibility retains 7,291 of 45,838 prepared clips (15.91%), increasing mean training duration from 2.183 to 4.404 seconds. Each eligible clip is presented six or seven times, whereas each selected Vietnamese clip appears once. Using the complete English pool in the equal-step F0 control gives 67.438 WER, worse than matched-pool FC by 0.620 points (95% interval [0.298, 0.950]). This alternative does not recover the gain; it changes content, duration and repetition together, and cannot show that additional data is intrinsically harmful.

The assessment durations are more similar: Vietnamese mean/median length is 145.74/129 frames, versus 151.31/139 in English. English's largest length stratum is nearly tied and the stratum above 200 frames is slightly worse with FM. The contrast therefore cannot be explained simply by shorter English test clips. The larger-data English reference reaches 36.010 WER, emphasizing the remaining gap to stronger training resources. Taken together, the English results bound the portability of FM and motivate studying exposure and supervision conditions, rather than attributing the difference to language alone.

5.2. Does the gain require fixed-time boundary supervision?

The matched augmentation follow-up directly compares fixed-time fragments with conventional whole-word views (Section 4.3). Table 5 shows the four matched states under the common decoder. We first compare the three crop-control states. Fixed-time mixing

Table 5: Source and destination performance of the matched continuation recipes at seed 17. The same fitted states use a common decoder on 11,307 VLR and 1,167 ViCocktail clips. Training frame and native-token counts are in millions.

Training views	VLR WER (%)	ViCocktail WER (%)	Frames	Tokens
Full-only	38.029	67.369	7.307	1.145
Fixed-time mixed	38.580	63.846	4.888	0.798
Whole-word mixed	38.606	63.388	5.013	0.784
Whole-word short-only	44.844	70.715	2.719	0.422

Each arm uses 48,000 source presentations and 6,000 updates. Both mixed arms share 317 full-clip fallbacks; short-only uses 598 fallbacks (1.25%). These fitted states are distinct from the original FC/FM comparison. Frame and token exposures are not equal.

reduces WER from 67.369 to 63.846, a 3.523-point gain with a conditional 95% group-bootstrap interval of [2.985, 4.179]. Whole-word mixing reaches 63.388 WER, improving over full-only by 3.981 points [3.420, 4.731].

Whole-word mixing is better than fixed-time mixing by 0.459 points [0.268, 0.674]. Its 138 fewer errors comprise 106 fewer deletions and 78 fewer insertions, offset by 46 additional substitutions. The transfer benefit thus persists with conventional word-aligned views, and the tested fixed-time rule does not provide an additional advantage over the tested whole-word alternative under this protocol. This supports mixed short supervision as a useful continuation choice here without requiring incomplete boundary evidence as its explanation.

The same three fitted states also expose the source cost of these transfer gains. On all 11,307 VLR test clips, WER is 38.029 for full-only, 38.580 for fixed-time and 38.606 for whole-word continuation. Relative to full-only, the source costs are 0.551 points [0.439, 0.664] for fixed-time and 0.577 points [0.448, 0.705] for whole-word mixing. Both arise primarily from more deletions, despite fewer substitutions. Whole-word source WER exceeds fixed-time WER by 0.026 points, with an interval of $[-0.043, 0.091]$. This comparison does not resolve a source-performance difference between the two crop rules or establish their equivalence. Whole-word mixing delivers the larger measured destination gain; neither mixed recipe improves uniformly over full-only continuation across source and destination.

The three crop-control arms receive the same source presentations and updates. Whole-word mixing nevertheless processes 2.55% more frames and 1.70% fewer target tokens than

Table 6: Predeclared comparisons with whole-word short-only continuation. Differences are reference minus short-only WER in percentage points; positive values favor short-only. Mixed denotes whole-word mixing.

Population	Contrast	Difference	95% interval
VLR	Mixed–short-only	-6.238	[-6.592, -5.887]
VLR	Full–short-only	-6.815	[-7.207, -6.440]
ViCocktail	Mixed–short-only	-7.328	[-8.864, -5.488]
ViCocktail	Full–short-only	-3.347	[-4.773, -1.341]

Pointwise intervals use 10,000 paired bootstrap resamples of 61 archived VLR source groups or 20 ViCocktail recording groups, conditional on the fitted states. Whole-word mixed versus short-only is the primary contrast; full-only versus short-only is the reference contrast.

fixed-time mixing. The comparison therefore evaluates two practical crop recipes rather than isolating a single boundary variable. It is a one-seed follow-up on the previously used assessment corpus, with shared fallback positions. The original FM source/destination profile below belongs to different fitted states; Table 5 reports the new controls' own source assessment.

5.3. Does the gain persist without scheduled full-clip supervision?

The short-only state obtains 70.715% ViCocktail WER and 44.844% VLR WER (Table 5). The predeclared whole-word mixed–short-only differences are -7.328 destination points $[-8.864, -5.488]$ and -6.238 source points $[-6.592, -5.887]$, favoring mixing in both populations. Short-only is also worse than full-only continuation by 3.347 destination points and 6.815 source points (Table 6).

The loss is dominated by deletions. Relative to whole-word mixing, short-only adds 5,735 deletions on ViCocktail; 3,098 fewer substitutions and 432 fewer insertions leave 2,205 additional errors. On VLR, 33,541 additional deletions outweigh 15,373 fewer substitutions and 4,072 fewer insertions, adding 14,096 errors. This error balance is consistent with poorer complete-transcript coverage, but does not identify its internal cause.

Retaining scheduled full-utterance presentations therefore improves the tested whole-word recipe at the same update budget. The benefit of mixing over full-only continuation cannot be extended to a nearly all-short schedule: the latter uses 47,402 short views and 598 full fallbacks, and performs worse than both controls. Frame and target-token

Table 7: Loss-reduction control on the complete ViCocktail assessment at seed 17. All four fitted states use the common decoder. Gain is full-only minus fixed-time WER.

Loss reduction	Full-only WER	Fixed-time WER	Gain (pp)
Native	67.369	63.846	3.523
Token-normalized	67.223	63.763	3.460

exposure also decrease, so this comparison does not isolate context length from objective exposure or establish an optimal mixing ratio.

5.4. Does the transfer gain persist under token-normalized loss?

The paired reduction control tests whether the fixed-time gain depends on retaining the native unnormalized components. Table 7 reports the complete four-state comparison. With token-normalized loss, full-only and fixed-time mixing obtain 67.223 and 63.763 WER. The gain is 3.460 points, with a conditional 95% recording-group interval of [2.756, 4.423].

The normalized-minus-native change in the gain is −0.063 points, with a paired interval of [−0.366, 0.386]. Thus the transfer advantage remains under the tested component-wise normalization. The interval does not establish exact equivalence between reductions, but the result does not support an explanation in which the gain disappears after dividing by target length. This one-seed control strengthens the short-view finding under a second objective reduction; it does not identify a unique optimization mechanism or predict how normalized whole-word mixing would behave.

5.5. Source and destination performance at the same fitted states

For the original continuation comparison, we assess the initializer, FC and FM on the complete prepared source test sets: 11,307 VLR clips and 1,243 LRS2 clips. This retrospective follow-up uses the same seed-17 checkpoints and readouts as the destination comparison, without source-test decoder tuning. Table 8 and Figure 4 show the resulting source/destination profiles.

On VLR, common-decoder WER is 39.830 for the initializer, 38.016 for FC and 38.604 for FM. The paired FC–FM difference is −0.588 points, with a 95% source-group bootstrap interval of [−0.709, −0.474]. FM therefore trades a modest source disadvantage relative

Table 8: Source and destination WER (%) at the same fitted states and common decoder. Source populations are 1,243 LRS2 clips (6,660 words) and 11,307 VLR clips (225,960 whitespace units). Destination populations are 2,854 WildVSR and 1,167 ViCocktail clips. Continuation results use seed 17; each corpus is scored separately. Lower WER is better.

	English		Vietnamese	
State	LRS2	WildVSR	VLR	ViCocktail
Initializer	56.577	67.092	39.830	70.034
FC	50.976	66.817	38.016	67.326
FM	52.432	66.884	38.604	63.916

to FC for its 3.410-point ViCocktail improvement at this seed. It adds 1,329 source errors: 3,641 additional deletions outweigh 2,163 fewer substitutions and 149 fewer insertions. Both continuations still improve on the initializer.

On LRS2, common-decoder WER is 56.577, 50.976 and 52.432 for the initializer, FC and FM. FM produces 97 more errors than FC: 32 substitutions, 26 deletions and 39 insertions across 6,660 reference words. The FC–FM difference is −1.456 points, with an interval of [−2.375, −0.584]. Unlike the Vietnamese profile, this source disadvantage has no demonstrated destination benefit on WildVSR. It is a disadvantage relative to FC, not forgetting relative to the initializer; the experiment does not measure the English initializer's upstream LRS3 performance.

Decoder selection changes the Vietnamese source ranking. Applying the frozen ViCocktail-development choices to VLR gives FC/FM WERs of 43.371/38.939 and an FC–FM difference of 4.432 points [4.199, 4.658]. FC uses greedy CTC with blank bias −1, whereas FM uses hybrid beam decoding with CTC weight 0.5. These are destination-calibrated readouts, not source-optimal decoders, and their contrast combines changes in the fitted model and decoding. The common decoder exposes the smaller source cost described above. The English source ranking is unchanged by its development-selected readouts: FC/FM WER is 50.976/52.883, a difference of −1.907 points [−2.986, −0.831].

The source follow-up separates a transfer benefit from uniform improvement across corpora. It also shows why source retention must be reported with its readout: a ranking obtained with destination-calibrated decoders need not match the ranking under shared

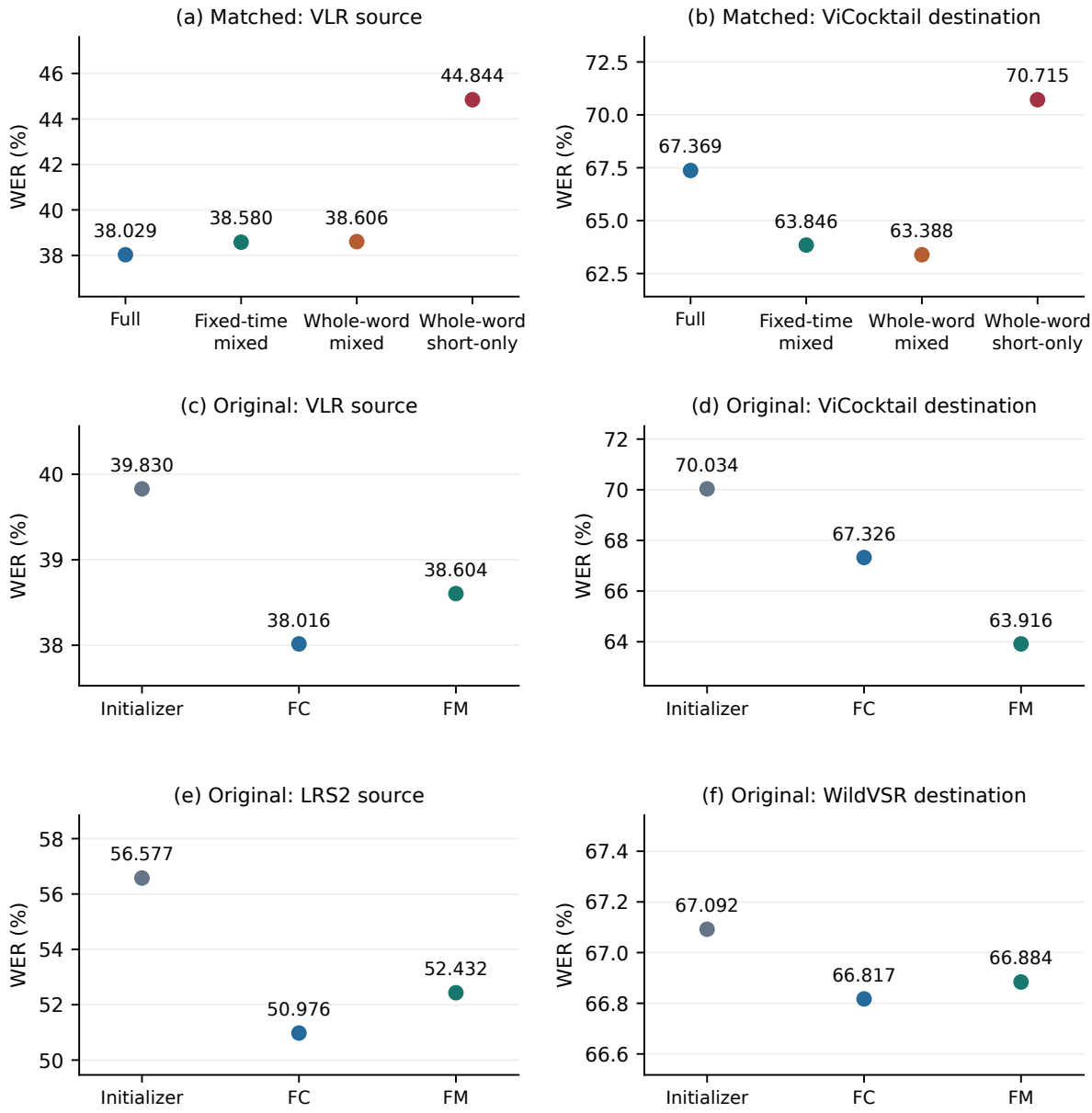


Figure 4: Source/destination WER at the same fitted seed-17 states under common decoding. (a,b) The matched full-only, fixed-time mixed, whole-word mixed and short-only states. (c,d) The original Vietnamese initializer/FC/FM comparison. (e,f) The original English comparison. Original and matched FC/FM states are distinct. Each panel has a separately labelled vertical scale; WERs are not averaged across corpora.

decoding. Neither profile identifies the learning mechanism responsible for the difference between continuation settings.

5.6. Error changes and utterance length

Under the common decoder, Vietnamese FM removes 1,026 errors at seed 17: 34 substitutions, 492 deletions and 500 insertions. At seed 29, it removes 954 errors: 35 substitutions, 371 deletions and 548 insertions (Figure 5). Most of the gain therefore

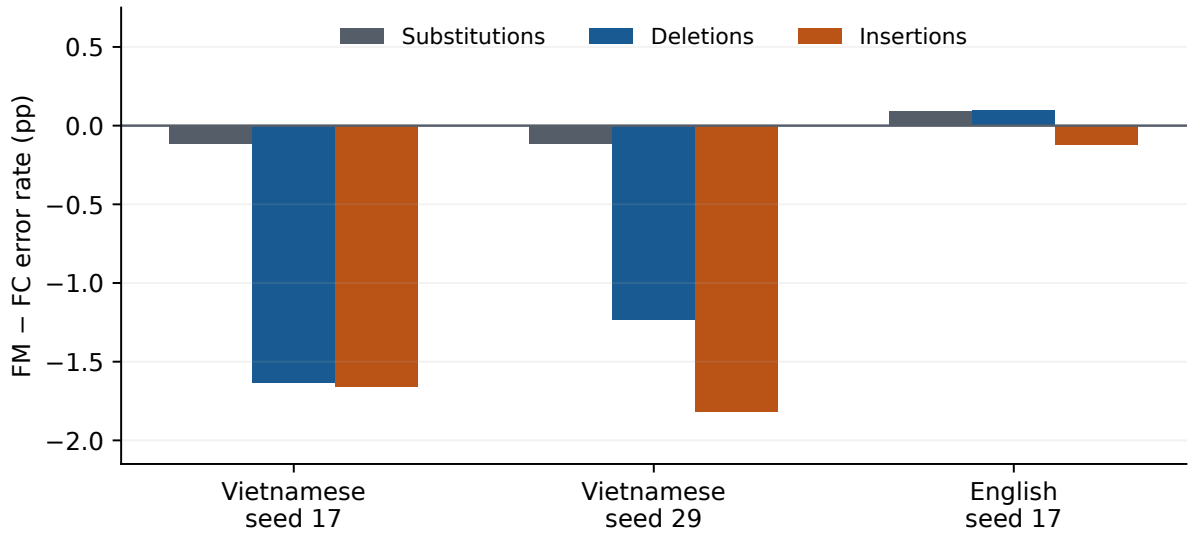


Figure 5: Common-decoder changes in substitution, deletion and insertion rates, computed as 100(FM count − FC count)/N. Negative values indicate fewer errors. Each setting uses its own complete assessment reference count.

comes from producing a more accurate number of transcript units, rather than substituting fewer units. The counts are consistent with a change in omission–insertion behavior; they do not identify whether the cause lies in visual representations, attention or confidence calibration.

A post-hoc length analysis shows that the Vietnamese gain extends beyond short utterances. The largest stratum, 101–200 prepared frames, contains 772 clips and 17,820 reference units. FM removes 398 and 349 errors at seeds 17 and 29, respectively, reducing WER by 2.233 and 1.958 points. Clips above 200 frames also improve, by 3.814 and 3.275 points. The 11 shortest clips contain only 57 reference units and account for 58 and 63 errors removed; their extreme percentage-point changes do not characterize the aggregate result. These descriptive strata were formed after inspecting the aggregate outcomes.

5.7. Training exposure and the role of context

FM changes the length of both the visual input and its supervision. Table 9 quantifies this change at the common final update. Vietnamese FM uses approximately 66.4% of FC's input frames and 69.1% of its target tokens in both seeds. Equal numbers of updates and presentations therefore do not imply equal information exposure. These counts do not measure FLOPs, runtime or gradient scale.

Table 9: Training exposure at 6,000 updates. Every run has 48,000 presentations; frame and native-token counts are shown in millions.

Setting	Seed	FC frames	FM frames	FC tokens	FM tokens
Vietnamese	17	7.307	4.851	1.145	0.791
Vietnamese	29	7.293	4.842	1.144	0.790
English	17	5.284	3.843	0.831	0.629

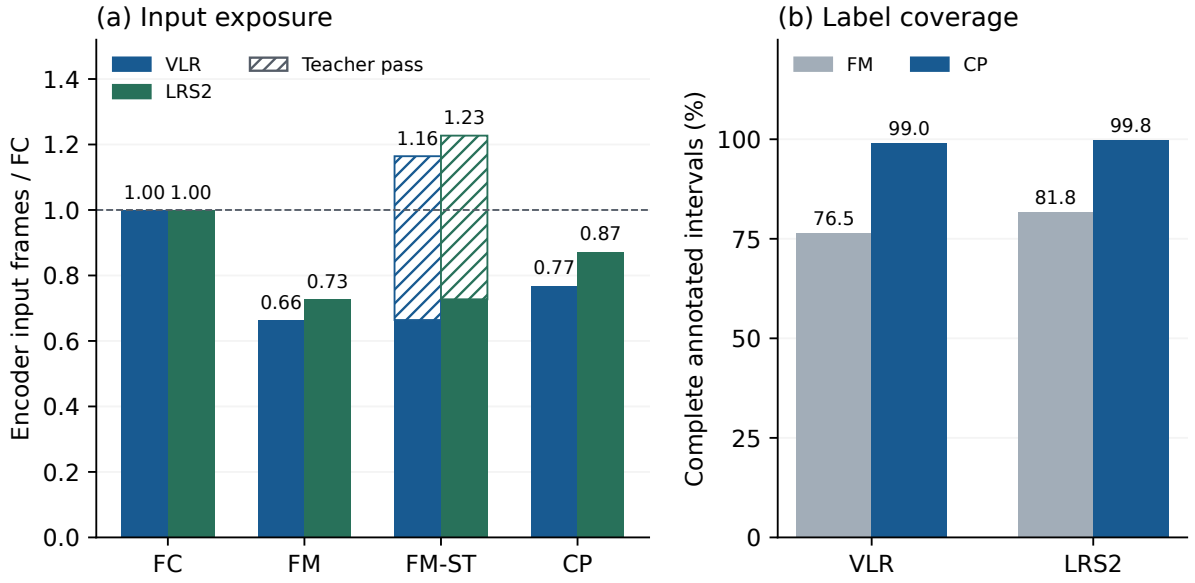


Figure 6: Training-design measurements at seed 17. (a) Encoder input frames relative to FC, including the no-gradient teacher pass in FM-ST. (b) Selected positive-duration annotation intervals fully contained in the encoder input. Coverage uses all 24,000 fragment presentations per corpus and automatic timestamps; zero-duration entries are excluded.

The two context controls retain FM's fragment targets. CP supplies neighboring pixels, adding 768,000 encoder input frames across 24,000 fragment presentations. FM-ST instead supplies a full-view predictive distribution through 24,000 additional teacher passes. At seed 17, total encoder input is 76.90% of FC for Vietnamese CP and 116.42% for FM-ST (Figure 6). Teacher passes do not backpropagate, so the ratios are exposure comparisons rather than compute-matched budgets.

Added context also restores visual evidence for labels touching a crop boundary. The fraction of selected positive-duration annotation intervals fully contained in the input rises from 76.49% to 99.02% in VLR, and from 81.79% to 99.77% in LRS2. A one-frame tolerance preserves this contrast. The 2,378 zero-duration VLR entries are excluded from coverage calculations but remain in training. Thus, CP changes neighboring information and boundary-label coverage together; it is not a pure intervention on context independent

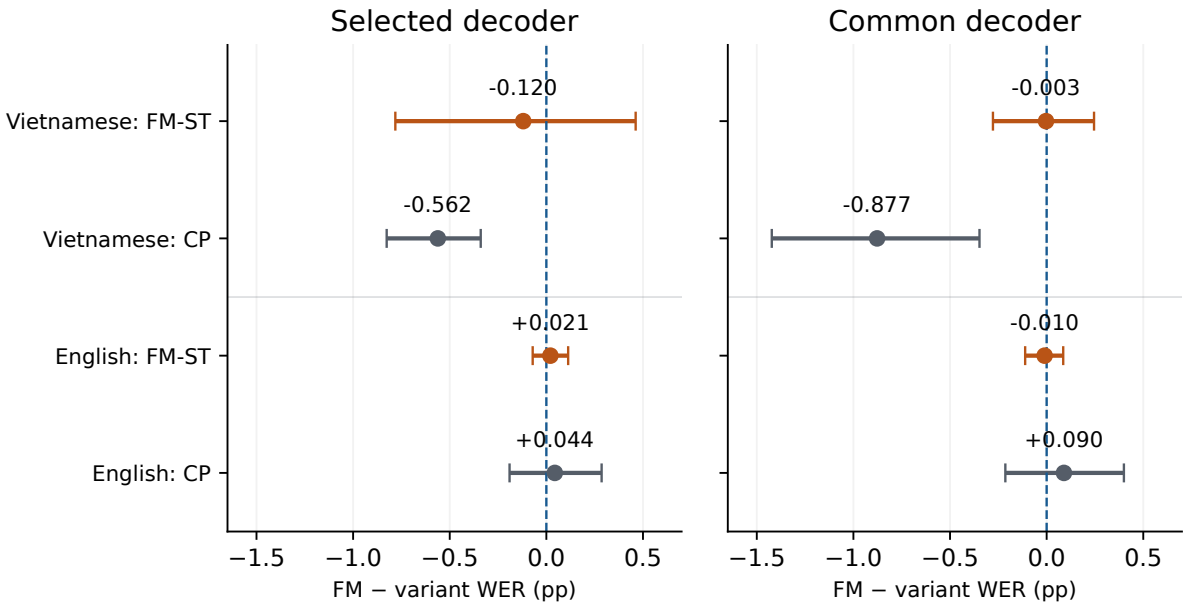


Figure 7: Paired effects of full-view self-teaching (FM-ST) and context preservation (CP) relative to FM at seed 17. Positive values favor the variant. Error bars are pointwise 95% bootstrap intervals, conditional on fitted models and decoders, without multiple-comparison adjustment. Vietnamese resampling uses original-video groups; English resampling uses released clips.

of label completeness. Automatic timings describe annotation coverage, not verified visual articulation.

Neither control strengthens the main Vietnamese result. FM-ST reaches 62.002 primary and 63.919 common-decoder WER, compared with FM's 61.882 and 63.916. The common-decoder difference is one error across 30,091 reference units; both paired intervals include zero. The tested self-teaching loss therefore adds no clear benefit at its specified temperature and weight.

CP reaches 62.444 primary and 64.793 common-decoder WER. It remains better than FC by 5.001 and 2.532 points, but FM outperforms CP by 0.562 points under the primary decoder (95% interval [0.340, 0.826]) and 0.877 points under the common decoder ([0.348, 1.422]). FM and CP select the same primary decoder. Restoring most boundary-label coverage preserves a substantial benefit over FC, while failing to improve FM. This supports further investigation of shorter supervision; it does not establish that missing context causes the gain.

CP's Vietnamese error balance differs from FM: 325 fewer substitutions and six fewer insertions are offset by 595 additional deletions, producing 264 more errors overall. This

post-hoc decomposition shows why restoring visual coverage or reducing one edit type need not improve complete-transcript WER.

In the English setting, FM-ST and CP remain near FM: their primary WERs are 66.771 and 66.749, respectively, and all English FM–variant intervals include zero. Under the common decoder, CP removes 621 substitutions and 461 insertions relative to FM but adds 1,039 deletions, leaving only 43 fewer errors (0.090 points). Likewise, common-decoder FM removes 59 insertions relative to FC but adds 44 substitutions and 47 deletions. The Vietnamese improvement in both insertion and deletion counts does not carry over to this English setting.

6. Discussion

6.1. When fragment supervision is useful

The original Vietnamese FC/FM gain repeats across both fitted seeds and all 20 observed recording groups, mainly through fewer deletions and insertions. It accompanies a source cost relative to FC, while FM remains better than the VLR initializer. The English comparison retains less source improvement than FC without a clear WildVSR benefit. Joint source/destination assessment therefore identifies both the Vietnamese transfer benefit and its regime dependence. Conventional whole-word mixing strengthens the Vietnamese transfer gain without resolving its source cost. The effect therefore does not require incomplete boundary evidence from fixed-time crops.

The short-only control directly tests the contribution of scheduled full supervision. Replacing full views with additional whole-word fragments raises destination and source WER by 7.328 and 6.238 points, respectively, relative to whole-word mixing. Additional deletions outweigh fewer substitutions and insertions. An exploratory fixed greedy-CTC check preserves this disadvantage against both full-only and whole-word mixed states on both complete populations (Supplementary Material, Tables 1 and 4). The ordering is thus consistent across the two tested readouts of the same fitted states. Retaining full presentations benefits this fixed-update schedule. Because short-only also reduces frame

and token exposure, the contrast evaluates the complete schedule rather than an isolated mixing effect at equal exposure.

This distinction matters when selecting a continuation strategy. Source recognition and destination recognition answer different questions, and their WERs should be considered separately. Source retention also depends on the readout: the Vietnamese FC/FM ranking reverses when destination-calibrated decoders replace the shared decoder. Reporting both comparisons distinguishes the fitted-state comparison from the deployed model-plus-decoder choice.

FM has a practical advantage in deployment complexity: it retains the same single whole-utterance recognition procedure. It is relevant when source videos and an existing recognizer are available but additional inference modules or destination-domain gradient updates are undesirable. The Vietnamese experiment does use destination development data for decoder calibration; the shared-decoder result separates the observed gain from model-specific calibration. The observed error rates motivate assisted transcription with human correction as a potential application; user-level utility remains to be evaluated.

6.2. What changes when context is restored

The context-preserving control retains the fragment targets while restoring neighboring visual evidence. It covers almost all selected annotation intervals and retains part of the Vietnamese gain over FC, but performs worse than FM. Its reduction in substitutions is outweighed by additional deletions. This same-target contrast shows that restoring visual context does not improve the tested continuation recipe. Since context and label coverage change together, the comparison characterizes the intervention as a whole.

Full-view self-teaching provides contextual information through a predictive distribution instead of additional student input. At the tested temperature and weight, it adds no clear benefit. The simpler FM recipe therefore retains the strongest transfer result among these tested context variants.

FM couples shorter inputs, shorter supervision and native loss reduction. The English setting also repeatedly revisits a smaller, length-selected source pool, whereas the Vietnamese setting presents each selected source clip once. These differences suggest that

supervision exposure and continuation regime deserve attention before attributing the contrast to linguistic structure. The whole-word comparison shows that the transfer benefit extends to conventional short-view augmentation, supporting temporal supervision as the relevant design choice. Crop extent and target exposure vary with the labelling rule, so the whole-word advantage reflects that combined change. The paired reduction control retains a 3.460-point fixed-time gain after normalizing each loss component by its valid target length, close to the native 3.523-point gain. The transfer advantage therefore persists under the tested alternative reduction. Together, these controls narrow the explanation beyond a particular boundary rule or the native loss reduction, while leaving internal optimization mechanisms open.

6.3. Scope and reproducibility

The evidence concerns two corpus pairs within one recognizer family. The Vietnamese FC/FM comparison includes two seeds; the English comparison, context controls and matched crop, short-only and loss controls use seed 17. The assessment sets have been used in earlier experiments, and the timing labels are automatic. Bootstrap intervals describe variation within the observed assessment populations conditional on fitted states; they do not estimate variation across all possible training runs. The measured frame/token counts describe exposure, not gradient contributions or compute efficiency. The tested schedules establish neither an optimal mixing ratio nor the behavior of other short-only objectives. Independent corpus confirmation and a second recognizer family would test the portability of the result more directly than additional analysis of the same test sets.

Reproducibility materials include frozen training and assessment plans, preparation, training and inference source, model identities, per-clip edit counts, and CPU scripts for numerical reproduction. Neural reruns additionally require the providers' data and checkpoints. The derived measurements and scientific source do not relicense third-party assets.

7. Conclusion

Fragment-mixed continuation changes training supervision while preserving ordinary whole-utterance visual speech recognition. Under matched source presentations and optimizer updates, it reduces VLR-to-ViCocktail WER by 3.41 and 3.17 points with a common decoder across two seeds. Conventional whole-word mixing also improves transfer and outperforms the tested fixed-time rule by 0.459 points. The fixed-time gain persists under token-normalized loss; restoring neighboring pixels or full-view predictive guidance does not improve the tested recipe.

The short-only control clarifies the schedule's contribution. Replacing the scheduled full views with more whole-word fragments worsens destination WER by 7.328 points and source WER by 6.238 points relative to whole-word mixing, primarily through additional deletions. Retaining full supervision is therefore beneficial in this fixed-budget comparison. Mixing still costs 0.55–0.58 source WER points relative to full-only continuation, and the English setting shows no clear WildVSR gain despite a source disadvantage relative to FC. These findings identify mixed temporal supervision as a useful continuation strategy in the studied Vietnamese regime. Joint assessment of source retention, destination recognition and actual supervision exposure shows the value of retaining complete utterances alongside fragments, and defines the trade-offs to assess when applying the strategy in another continuation setting.

References

[1] B. Shillingford, Y. Assael, M. W. Hoffman, T. Paine, C. Hughes, U. Prabhu, H. Liao, H. Sak, K. Rao, L. Bennett, M. Mulville, M. Denil, B. Coppin, B. Laurie, A. Senior, N. de Freitas, Large-scale visual speech recognition, in: Proceedings of Interspeech, 2019, pp. 4135–4139. doi:10.21437/Interspeech.2019-1669.
URL https://www.isca-archive.org/interspeech_2019/shillingford19_interspeech.html

[2] P. Ma, A. Haliassos, A. Fernandez-Lopez, H. Chen, S. Petridis, M. Pantic, Auto-AVSR: Audio-visual speech recognition with automatic labels, in: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023, pp. 1–5. doi:10.1109/ICASSP49357.2023.10096889.
URL https://arxiv.org/abs/2303.14307

[3] Y. A. D. Djilali, S. Narayan, E. LeBihan, H. Boussaid, E. Almazrouei, M. Debbah, Do vsr models generalize beyond lrs3?, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2024, pp. 6635–6644.
URL https://arxiv.org/abs/2311.14063

[4] B. Martinez, P. Ma, S. Petridis, M. Pantic, Lipreading using temporal convolutional networks, in: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020, pp. 6319–6323. doi:10.1109/ICASSP40776.2020.9053841.
URL https://arxiv.org/abs/2001.08702

[5] K. R. Prajwal, T. Afouras, A. Zisserman, Sub-word level lip reading with visual attention, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 5162–5172.
URL https://arxiv.org/abs/2110.07603

[6] J. S. Chung, A. Senior, O. Vinyals, A. Zisserman, Lip reading sentences in the wild, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017. doi:10.1109/CVPR.2017.367.
URL https://robots.ox.ac.uk/~vgg/publications/2017/Chung17/

[7] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, A. Zisserman, Deep audio-visual speech recognition, IEEE Transactions on Pattern Analysis and Machine Intelligence 44 (12) (2022) 8717–8727, first available online in 2018; journal issue published in 2022. doi:10.1109/TPAMI.2018.2889052.
URL https://arxiv.org/abs/1809.02108

[8] B. Shi, W.-N. Hsu, K. Lakhotia, A. Mohamed, Learning audio-visual speech representation by masked multimodal cluster prediction, in: International Conference on Learning Representations, 2022.
URL https://arxiv.org/abs/2201.02184

[9] A. Graves, S. Fernández, F. Gomez, J. Schmidhuber, Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks, in: Proceedings of the 23rd International Conference on Machine Learning, ACM, 2006, pp. 369–376. doi:10.1145/1143844.1143891.
URL https://www.cs.toronto.edu/~graves/icml_2006.pdf

[10] S. Watanabe, T. Hori, S. Kim, J. R. Hershey, T. Hayashi, Hybrid CTC/attention architecture for end-to-end speech recognition, IEEE Journal of Selected Topics in Signal Processing 11 (8) (2017) 1240–1253. doi:10.1109/JSTSP.2017.2763455.
URL https://www.merl.com/publications/TR2017-190

[11] J. Xue, X. Liu, X. Wu, F. Yu, J. Wang, InfoSyncNet: Information synchronization temporal convolutional network for visual speech recognition, in: International Joint Conference on Neural Networks (IJCNN), 2025, pp. 1–8. doi:10.1109/IJCNN64981.2025.11228396.
URL https://arxiv.org/abs/2508.02460

[12] A. Baevski, W.-N. Hsu, Q. Xu, A. Babu, J. Gu, M. Auli, data2vec: A general framework for self-supervised learning in speech, vision and language, in: Proceedings of the 39th International Conference on Machine Learning, Vol. 162 of Proceedings of Machine Learning Research, PMLR, 2022, pp. 1298–1312.
URL https://proceedings.mlr.press/v162/baevski22a.html

[13] Z. Yao, W. Kang, X. Yang, F. Kuang, L. Guo, H. Zhu, Z. Jin, Z. Li, L. Lin, D. Povey, CR-CTC: Consistency regularization on CTC for improved speech recognition, in: International Conference on Learning Representations, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/432540ba4d35a2010ebfbaa3d5cb4640-Paper-Conference.pdf

[14] M. Kim, H.-I. Kim, Y. M. Ro, Prompt tuning of deep neural networks for speaker-adaptive visual speech recognition, IEEE Transactions on Pattern Analysis and Machine Intelligence 47 (2) (2025) 1042–1055. doi:10.1109/TPAMI.2024.3484658.
URL https://arxiv.org/abs/2302.08102

[15] T.-B. Nguyen, T. V. Nguyen, Q. T. Do, C. M. Luong, ViCocktail: Automated multi-modal data collection for vietnamese audio-visual speech recognition, in: Interspeech, 2025, pp. 166–170. doi:10.21437/Interspeech.2025-1559.
URL https://www.isca-archive.org/interspeech_2025/nguyen25d_interspeech.html

[16] Nguyen Quoc Trung, Pham Thi Thao Nguyen, Nguyen Ngoc Gia Ninh, Dam Nguyen Quynh Hoa, Matej Sindelar, Jakub Stefansky, Truong Hoang Vinh, Petr Bilik, Simon Bilik, Radek Martinek, A large-scale vietnamese lip-reading dataset for visual speech recognition, Zenodo, version 2, published 23 April 2026; version-specific record (2026). doi:10.5281/zenodo.19705466.
URL https://zenodo.org/records/19705466

[17] T. Afouras, J. S. Chung, A. Zisserman, LRS3-TED: A large-scale dataset for visual speech recognition (2018). arXiv:1809.00496, doi:10.48550/arXiv.1809.00496.
URL https://arxiv.org/abs/1809.00496